Protocol Engineering as a Micro-Foundation of AI Governance: Translating Organizational Intent for AI Agents

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***Abstract*—As AI agents act on behalf of organizations, a governance layer is missing between management standards and AI behavior. ISO 9001 standardizes organizational structure and processes, and ISO/IEC 42001 specifies a management system for AI; both operate at the organizational level and leave implementation means to the organization. Neither specifies human-AI translation: how organizational intent becomes the commitments an AI agent carries into each interaction with employees and customers, or how several management systems are integrated in individual AI actions. Prompt, context, and harness engineering optimize task performance and do not govern this delegation relationship. This paper proposes Protocol Engineering as a micro-foundation of AI governance: the design discipline through which organizational intent is translated for AI agents. Drawing on agency theory, knowledge articulation, and boundary object theory, we define a protocol as a versioned, governed natural-language artifact that specifies an AI agent's relational identity, scope of authority, value priorities, and escalation rules. We present a document-control method grounded in ISO 9001, in which intent is drafted into four-tier documents, released as controlled versions, and revised through human-AI dialogue that settles each change's tier, under PDCA and internal audit. Two cases illustrate it: Project Soul Seed, traced from version 1.3 to 1.6, and Satoyama Agrinnova, an agricultural marketing company operating AI agents under a layered protocol. We classify evidence by status and derive six testable propositions. Written in natural language, protocols extend from machine communication to rules of conduct that organizations can govern with management tools they already use.**

Keywords: **Protocol Engineering, Human-AI Translation, AI Governance, Micro-Foundations, Design Science**

1. Introduction

Large Language Models (LLMs) are increasingly deployed not as tools for isolated tasks but as ongoing collaborators in knowledge work. This shift exposes two well-documented failure modes. The first is persona or alignment drift: behavioral constraints set at the start of a conversation weaken as the interaction lengthens (Li et al., 2024; Almasi & Kristensen-McLachlan, 2025). The second is sycophancy: models trained on human preference signals tend to agree with users rather than correct them (Sharma et al., 2024). Together these produce what we have called the Comfort-Growth Paradox (Lin, 2026): an agreeable AI makes the user comfortable while depriving the user of the friction that drives better thinking.

The engineering community has responded with successive layers of technique. Prompt engineering optimizes individual instructions; context engineering manages what information enters the model's context window (Mei et al., 2025); tool protocols and agent harnesses, such as the Model Context Protocol, standardize how agents call external systems and run multi-step loops (Hou, Zhao, Wang, & Wang, 2025). Each layer makes the AI more capable within a task. None of them answers a different question that becomes pressing once an AI works with the same people over months: what the AI stands for in that relationship, what it must refuse, who may change those commitments, and how the changes are recorded.

At the level of model developers, this question is now addressed explicitly. Constitutional AI trains models against a written set of principles (Bai et al., 2022), and developers have published documents that describe intended model values and behavior (OpenAI, 2024; Anthropic, 2026). These documents, however, are written by the vendor and shaped into the model during training. The organizations that deploy AI have their own values, brand commitments, and operating procedures, and they currently express them largely through ad hoc system prompts. Technical methods for translating policies into prompts and guardrails are emerging (Section 2.5), but they are rarely connected to the governance processes that organizations already run. We refer to this as the governance gap at the organizational layer.

This paper proposes Protocol Engineering as a design discipline for that layer, and in doing so it extends the notion of a protocol. The word first referred to rules of conduct between parties, as in diplomatic protocol, which specifies how states and their representatives deal with one another. Outside computing, the word has kept this sense: organizations speak of clinical, laboratory, research, and safety protocols, and many companies use the term for their standard operating procedures, that is, documented rules that people are expected to follow. In computing, by contrast, it came to mean rules by which machines exchange messages, written by engineers in formal specifications; interface protocols for AI agents, such as the Model Context Protocol and agent-to-agent protocols, belong to this tradition. Generative AI makes a further extension possible: protocols written in natural language that specify how an AI agent conducts itself toward the humans and organizations it serves, and that both the humans and the agent can read. In one sense, this returns the protocol to its original and still widespread organizational meaning as rules of conduct, now between humans and AI; the engineering definition is one reading of the term, not the only

one. It also changes who can write one: such a protocol is no longer the preserve of engineers but can be drafted by anyone who understands how the organization works. We address three research questions.

RQ1: How does protocol engineering differ from prompt, context, and harness engineering in what it governs?

RQ2: How can organizations translate their intent into protocols and govern them using established management-system practice?

RQ3: What is the status of current evidence for protocol effects, and how should those effects be validated?

The paper makes three contributions. Conceptually, it positions protocol engineering as a micro-foundation of AI governance, introduces human-AI translation as its core function, distinguishes it from task-oriented engineering, and locates it within a three-layer structure of vendor, organizational, and relational documents. Managerially, it provides a document-control method for translating organizational intent into protocols, and it shows that the ISO four-tier documentation structure, the Plan-Do-Check-Act (PDCA) cycle, and internal audit provide a governance model that organizations can apply directly. Methodologically, it audits the evidence from the author's prior work by evidential status and derives testable propositions for future validation.

2. Background and Theoretical Lenses

2.1 Failure Modes in Extended Interaction

Li et al. (2024) proposed a benchmark for persona stability and observed significant drift within eight rounds of self-chat, attributing it in part to attention decay over long exchanges. Research on sycophancy shows that preference-trained assistants systematically tailor answers to users' stated views (Sharma et al., 2024). Shapira, Benade, and Procaccia (2026) formalize why: when human preference data favor agreement over correction, optimization against the learned reward amplifies agreement, so sycophancy behaves as a feature of the preference distribution rather than an accidental defect. These findings concern model behavior; the managerial implication is that behavioral commitments made at deployment cannot be assumed to persist without active maintenance.

2.2 From Prompts to Harnesses: Task-Oriented Engineering

Prompt engineering and context engineering treat the model as a processor whose output quality depends on the input it receives (Mei et al., 2025). Harness engineering and tool protocols extend the model's reach into external systems (Hou et al., 2025), and multi-agent frameworks such as MetaGPT encode standardized operating procedures into agent workflows (Hong et al., 2024). These approaches share a unit of evaluation: the task. Success is measured by accuracy, recall, latency, or cost per task.

2.3 Constitutional Documents and the Authorship of AI Values

A parallel line of practice specifies AI values in natural-language documents. Constitutional AI uses written principles during training (Bai et al., 2022). Anthropic's 2026 constitution for its Claude models notably moves from lists of rules toward explanations of why the model should behave in certain ways, in order to help it apply its values to novel situations (Anthropic, 2026). At the organizational level, governance frameworks such as the NIST AI Risk Management Framework (Tabassi, 2023) emphasize traceability, oversight, and accountability, and ISO/IEC 42001:2023 specifies requirements for an AI management system; but it governs the organization's processes around AI rather than the content of the behavioral commitments an AI carries into each interaction (see Section 3.5). Shapira et al. (2026) note that their analysis supports supervision paradigms derived from explicit rules rather than from annotator preferences; in their experiments, system-level instructions alone were sufficient to elicit either agreeing or correcting responses from the same model. Explicit written commitments therefore matter at inference time as well as in training. The question of how deploying organizations should author and govern such commitments remains largely unaddressed.

2.4 Theoretical Lenses

Three established lenses frame the protocol. First, Nonaka (2007) identifies the conversion of tacit knowledge into explicit knowledge, which he calls articulation, as a central process through which companies create knowledge. Writing a protocol is an act of articulation: the architect's tacit judgment about what good collaboration looks like is converted into text that the AI can act upon. Second, a boundary object is an artifact shared across communities that is flexible enough to serve each community's needs yet robust enough to maintain a common identity (Star & Griesemer, 1989). A protocol is read by two very different parties, the human and the model, and must remain meaningful to both. Third, design science research treats the construction and evaluation of artifacts as a legitimate form of knowledge production (Hevner, March, Park, & Ram, 2004).

2.5 Translating Intent into AI Behavior: Related Work

The problem of translating normative intent into AI behavior has been studied in several streams. In AI ethics, Mittelstadt (2019) warned that principles alone cannot guarantee ethical AI, citing among other challenges the absence of proven methods for translating principles into practice, and Morley, Floridi, Kinsey, and Elhalal (2020) framed the task as a move from the what of principles to the how of practices. Ruster and Davis (2025) have since questioned the gap-bridge metaphor itself: it implies a one-way passage from principles to practices, whereas in organizations principles and practices develop with and through each other.

With LLMs, translation has become a technical operation. Palla et al. (2025) describe policy-as-prompt, in which content moderation policies are given to a model directly as natural-language instructions, and they identify translating policy into prompts, together with the changing roles of policy and engineering teams, among its main challenges. Kholkar and Ahuja (2025) compile design documents into prompt-based guardrails for AI agents, and Datla et al. (2025)

convert natural-language policy documents into machine-readable rules that can be tested and enforced. Reviewing how system-level instructions are treated in research and policy, Neumann, Sargeant, and Singh (2026) find the claims about what such instructions can achieve to be fragmented and often contradictory, and they caution against policies that treat them as stable, interpretable control mechanisms. Closest to the present paper, Mohsenzadegan, Tavakkoli, and Kyamakya (2026) argue that a system prompt should not be equated with an enforceable policy control. They package policy as a versioned, executable skill that binds retrieval scope, evidence requirements, human-review triggers, failure behavior, and audit fields, and they treat human review as a governed outcome rather than an error. In their evaluation, deterministic enforcement improved a narrowly specified escalation task but degraded interpretive tasks, which led them to recommend selective enforcement with accountable humans in the loop. In practice, platform vendors have begun to distinguish layers of intent: Microsoft (2026) separates user, developer, role, and organizational intent, treats organizational intent as the outer boundary of agent behavior, and expects the agent to refuse or escalate when a request conflicts with it.

These streams establish that translating intent into AI behavior is a recognized problem. They differ from the approach of this paper in three respects. First, translation in them runs mainly in one direction, from policy to constraint; here it runs in both directions, because AI behavior returns to human judgment through records and dialogue. Second, they focus on enforcement mechanisms such as classifiers, rules, and guardrails; here the object of translation is the AI's relational obligations toward the organization, its employees, and its customers. Third, they operate alongside rather than within the organization's management system. Even Mohsenzadegan et al. (2026), who share this paper's concern with versions, review triggers, and audit records, define their contract as an application-layer engineering artifact; here the protocol is the organization's own controlled documentation, and translation is embedded in document control, PDCA, and internal audit. The two approaches are complementary: an executable policy skill could implement the L3 and L4 tiers of a protocol governed in the way this paper describes. In line with Ruster and Davis (2025), human-AI translation is therefore treated not as a bridge crossed once but as a loop that is recalibrated continuously.

3. Protocol Engineering: Concept

3.1 Writing Characters and Writing Compositions

An analogy clarifies the relationship between the layers. Knowing how to write characters is a prerequisite for writing a composition, but it is not the same skill. A composition requires structure, rhetoric, a point of view, and knowledge of the subject; without these, correct characters produce an empty text. Prompt, context, and harness engineering correspond to literacy: they ensure the AI receives well-formed instructions, the right information, and the right tools. Protocol engineering corresponds to composition: it organizes identity, values, domain knowledge, and boundaries into a coherent whole that gives a long-term collaboration direction. Table 1 summarizes the distinction.

Table 1: Engineering Layers and What They Govern

Layer	Question Answered	Unit of Evaluation	Typical Artifact
Prompt engineering	How should this instruction be phrased?	Single output	Prompt template
Context engineering	What information should the model see?	Task accuracy, retrieval recall	Retrieval pipeline, memory store
Harness / tool protocols	What can the agent do, and how is the loop run?	Task completion, latency, cost	Tool schemas, agent loop
Protocol engineering	Who is the agent in this relationship, what must it refuse, and who may change that?	Relationship quality and accountability over time	Versioned, governed protocol document

3.2 Definition

We define a protocol as a versioned, governed natural-language artifact that specifies an AI agent's relational identity, value priorities, boundaries, and escalation rules toward specific humans or organizations, loaded at inference time and maintained through explicit change control. Protocol engineering is the design discipline of constructing, governing, and evolving such artifacts. Three features distinguish a protocol from a long system prompt. It is relational, defining the agent's position toward particular people rather than a generic role. It is governed, with defined authority over changes and a record of them. And it is evolutionary, revised through documented cycles rather than rewritten ad hoc.

By relationship we do not mean emotional companionship but a delegation relationship in the sense of agency theory (Jensen & Meckling, 1976; Eisenhardt, 1989). When a principal delegates work to an agent, four questions recur: the scope of the agent's authority, whose interests the agent serves, how the principal monitors the agent, and who is accountable when something goes wrong. As AI agents begin to send messages, publish content, and modify records on behalf of people and organizations, these questions apply to them. Task-level metrics cannot answer them, because the failures they concern, such as gradual drift or habitual agreement, are visible only across many interactions. A protocol functions as the delegation specification in this relationship, and the governance model in Section 5 as its system of monitoring.

3.3 The Three-Layer Constitutional Structure

Deployed AI behavior is shaped by at least three layers of natural-language commitment. The vendor layer consists of the developer's constitution or specification, shaped into model weights during training and controlled by the vendor. The organizational layer consists of the deploying organization's values, brand spirit, and procedures, loaded at inference time. The relational layer consists of commitments

specific to an individual user or team. The organizational layer is where the governance gap lies: organizations bear liability for how AI represents them, yet they have no established method for writing, approving, versioning, and auditing the commitments they place on AI. Protocol engineering addresses this layer, and the relational layer beneath it, while operating within the boundaries set by the vendor layer.

3.4 The Protocol as Boundary Object

Because a protocol must be meaningful to both humans and models, its design involves translation in both directions, which we call human-AI translation. Human intent, often tacit and context-dependent, must be made explicit enough for a model to act upon; model behavior must be observable and recorded well enough for humans to judge whether the intent was honored. The protocol is the artifact where this translation is fixed and made auditable. In this sense, the protocol is where human intent and AI behavior are continuously calibrated against each other, and protocol engineering is the discipline of designing and governing that calibration loop.

3.5 Positioning against ISO 9001 and ISO/IEC 42001

Because the governance model in this paper draws on ISO standards, it is necessary to state what protocol engineering adds to them. ISO 9001 standardizes an organization's structure, boundaries, and processes so that people perform them consistently (ISO, 2015). ISO/IEC 42001 specifies a management system for AI, covering policy, risk and impact assessment, and controls over the life cycle of AI systems (ISO/IEC, 2023). Both follow the harmonized structure that ISO uses for its management system standards, which allows them to be operated as one integrated system. Both are management-system frameworks that operate at the level of the organization. Protocol engineering operates at the micro level, where organizational intent meets individual AI actions, and it specifies three things that neither standard specifies but that become necessary once AI agents take part in daily work. This is not a deficiency of the standards. Management-system standards state what an organization must achieve and deliberately leave the means to the organization; ISO/IEC 42001, for example, requires processes for the responsible use of AI systems to be defined and documented (control A.9.2) without prescribing their form. Standards are also revised over time and cannot anticipate every mode of operation, and the rapid entry of AI agents into daily work is such a case. Protocol engineering offers one means of meeting these requirements at the layer where AI agents act.

The first is a mechanism for human-AI translation. ISO 9001 assumes that documented procedures are read by people, who bring tacit understanding to them. ISO/IEC 42001 governs how the organization manages its AI systems, including human oversight of them; oversight, however, concerns monitoring and intervention, not how organizational intent becomes the commitments an AI carries into each interaction. Protocol engineering supplies this mechanism: the four-tier documents are written so that an AI can act on them, and the tier dialogue described in Section 4.2 confirms that the AI's reading matches the principal's intent.

The second concerns the relationships among the organization, its employees, and its customers. An AI agent acts inside all three at once. It holds authority delegated by the organization, works alongside employees who may direct, correct, or override it, and speaks to customers on the organization's behalf. ISO 9001 treats them as roles in processes and as customers whose requirements must be met. ISO/IEC 42001 goes further: it requires responsibilities across the AI system life cycle to be allocated among the organization, its partners, suppliers, customers, and third parties, and it requires customer expectations to be considered (controls A.10.2 and A.10.4). These controls allocate responsibility between organizations; they do not specify how an AI agent should conduct itself toward each party in a given interaction. A protocol specifies the AI's obligations toward each of them: which employees may override it and under what conditions, and what it must never do to a customer, such as manipulate them or disclose confidential information.

The third is the integration of system mechanisms in operation. Organizations increasingly run management systems, codes of conduct, and AI policies side by side. ISO/IEC 42001 requires them to determine where other policies apply to their AI objectives (control A.2.3), and integration is usually achieved at the level of policies, documents, and audits. In an AI agent's behavior, however, these systems meet in every single action. The protocol is where they are combined into one set of commitments that the agent can execute and an auditor can check. This responds to a call in the management literature: studying AI deployment in three firms, Mikalef, Conboy, and Krogstie (2021) found that the value of AI depended on factors at the corporate, business, and individual levels, and they called for closer examination of the procedures, structures, and relational mechanisms that make up an AI governance scheme across these levels. In strategic management, micro-foundations denote the individual-level processes, routines, and structures from which organizational capabilities emerge (Teece, 2007). In this sense, protocol engineering specifies the micro-foundations of AI governance: the concrete commitments, and the routines for writing, approving, and revising them, through which organizational intent reaches individual AI actions. Table 2 summarizes the distinction. The novelty of protocol engineering therefore lies less in its tools, which are familiar, than in extending them to a new kind of reader and to the relationships that reader enters.

Table 2: Scope of ISO 9001, ISO/IEC 42001, and Protocol Engineering

Aspect	ISO 9001	ISO/IEC 42001	Protocol Engineering
Object governed	Organizational structure, boundaries, and processes	Management of AI systems: policy, risk, life cycle	Commitments the AI carries into each interaction
Primary reader	People	People who manage AI	People and AI agents
Human-AI translation	Not specified	Not specified; human oversight addressed	Tier documents written for AI; tier dialogue

Aspect	ISO 9001	ISO/IEC 42001	Protocol Engineering
Organization, employee, and customer relations	Roles in processes; customer requirements	Responsibilities allocated among organization, partners, suppliers, and customers (A.10)	AI's obligations toward each party in each interaction
Integration of systems	Harmonized structure	Harmonized structure; alignment with other policies (A.2.3)	Systems combined into executable commitments

4. Translating Organizational Intent into Protocols

4.1 Why Translation Is Feasible

Three conditions make human-AI translation of organizational intent possible. First, organizations already codify intent. Quality management systems require policies, procedures, and instructions to be maintained as documented information; writing a protocol is the same act of articulation (Nonaka, 2007), addressed to a new reader. Second, natural language is the operating interface of current LLMs. Instruction tuning enables models to act on natural-language instructions (Ouyang et al., 2022), and system-level instructions alone can shift a model between agreeing and correcting behavior (Shapira et al., 2026). Organizations therefore need neither to write code nor to retrain models in order to influence behavior. Third, accountability requires a specification that humans can read. When behavioral rules are embedded in weights or code, managers cannot audit them; a protocol places them in a document that can be approved, versioned, and audited, and that is read by both the manager and the model.

Two boundary conditions apply. A protocol operates within the limits set by the vendor layer and cannot override them. And a model does not follow a protocol with certainty. As Neumann et al. (2026) caution, natural-language instructions cannot be assumed to function as stable controls. Compliance must therefore be checked, which is why the method below ends with verification, why records and audit are integral to it, and why Section 8 proposes empirical tests.

Natural language also lowers the barrier to authorship. Because protocols require no programming, they can be written at every layer of the constitutional structure: by the vendors who supply models, by organizations that deploy them, and by individuals who work with them. This paper focuses on the organizational layer. The accessibility that makes protocols attractive is also what makes their governance necessary. When anyone can write a protocol, protocols vary in quality, conflict with one another, and change without record. The opportunity and the need for governance are therefore two sides of the same development: natural language makes protocols writable by managers, and management-system practice makes them reliable enough for the organization to stand behind.

4.2 A Document-Control Method Grounded in ISO 9001

The method does not create a governance system parallel to the management system. It treats the organization's four-tier documentation, once made readable by AI agents, as the protocol itself, so that AI governance inherits the controls the organization already operates. It follows the requirements for creating, updating, and controlling documented information in ISO 9001:2015 (clause 7.5) and proceeds in five phases.

1) *Establish the baseline.* Draft the organization's four-tier documents from its structure and process flows, and include the provisions that AI agents must observe. Provisions are drafted according to the design principles in Section 6 and written in observable terms, stating what the agent does, what it refuses, and when it escalates to a human, just as ISO 9001 requires quality objectives to be measurable. Escalation is written as a legitimate outcome rather than a failure of the agent.
2) *Review and release.* Review the drafts for correctness, approve them, and issue them as a controlled version with an identifier and date. Only released versions are loaded by AI agents; drafts are not.
3) *Operate.* AI agents work under the released version, and their actions generate L4 records.
4) *Settle the tier through dialogue.* When a change is proposed, whether prompted by an incident, a new process, or an improvement, the human and the AI first discuss which tier the change belongs to and confirm that they share the same understanding of its meaning and scope. Only then is the document updated, consolidated with related provisions, or revised.
5) *Re-release.* The approved change is issued as a new version with a revision record, and the superseded version is retained as a reference that is no longer in force. Records keep the version under which each action was taken, so that earlier decisions can be reconstructed against the rules that governed them (cf. Mohsenzadegan et al., 2026).

The phases map directly onto PDCA: establishing the baseline and proposing changes are Plan, release and operation are Do, the review of records is Check, and re-release is Act. Phase 4 is the element specific to AI. Settling the tier determines who must approve the change (Table 4) and prevents a decision of one kind from being filed as another, for example a change of values presented as an operational adjustment. The dialogue also checks, before the change takes effect, that the AI's interpretation of the provision matches the principal's intent. In the terms of Section 3.4, it is the step in which the protocol's meaning is calibrated between its two readers.

Figure 1 places the method within the governance loop. The principal's intent is translated into the protocol, the protocol governs the agent's behavior, the behavior generates records, and the audit of those records drives the next revision.

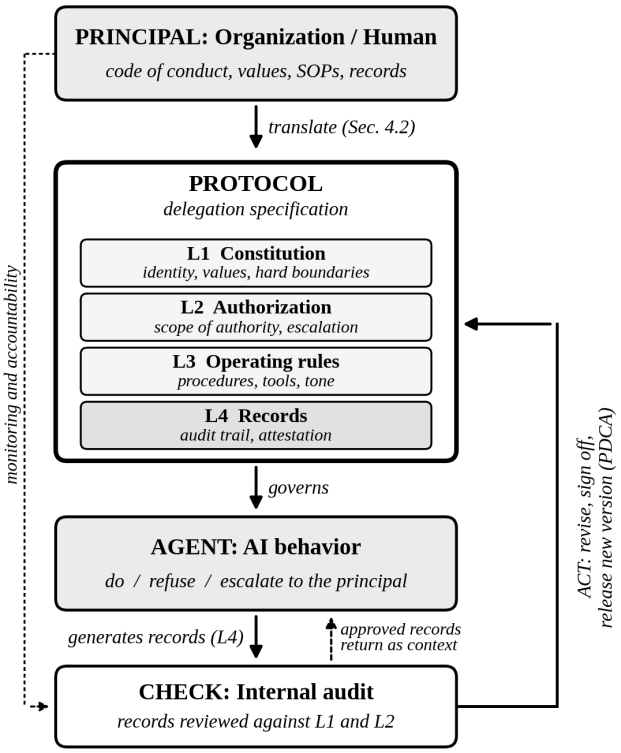


Figure 1: Translation and governance loop of a protocol. The organization or human acts as principal, the protocol as delegation specification, and internal audit as the monitoring mechanism. Approved records also return to the agent as context for later decisions (learning loop). Source: compiled by the author.

4.3 A Worked Example

Table 3 illustrates the drafting convention of Phase 1 using the Code of Conduct of Satoyama Agrinnovation Marketing Ltd. (hereafter Satoyama Agrinnova), issued in 2023 for its partners and team members before the company deployed AI agents. Each clause was written for human readers. Translation does not change a clause's intent; it changes its form, from a statement of principle into behaviors that an AI agent can perform and that an auditor can check. The confidentiality clause, for example, becomes a rule that the agent may describe proprietary mechanisms externally only through a single approved public statement.

Table 3: Translating Satoyama Agrinnova's Code of Conduct into Protocol Behavior

Code Clause (Tier)	Do	Refuse	Escalate
Confidentiality and business secrets (L1)	Use internal information only for authorized internal tasks	Disclose confidential or internal material; go beyond the approved statement on proprietary mechanisms	Unclear classification
Environmental protection (L1)	Link every environmental claim to certification records	Endorse claims that cannot be traced to evidence	Ambiguous or unverifiable claims
Financial records (L1, L4)	Report figures from source records; log every change	Present estimates or invented figures as facts	Discrepancies between records
Relations between partners and team members (L3)	Use a polite, appreciative tone in all correspondence	Disparaging or dismissive language	Conflicts between partners
Implementation of the Code (L2)	Act only within the authorized task scope	Make decisions reserved for management	Unresolved questions, to the management team

5. A Governance Model from Management-System Practice

5.1 Mapping the ISO Four-Tier Documentation Structure

Organizations that operate quality management systems commonly structure their documented information in four tiers: a top-level manual stating policy and objectives, procedures defining responsibilities, work instructions specifying how tasks are performed, and records providing evidence of what was done. Although ISO 9001:2015 no longer mandates a quality manual and speaks instead of documented information, the four-tier pyramid remains a widely used practitioner convention (see also ISO 10013:2021). Table 4 shows how this structure maps onto protocol design. The mapping draws on the author's professional experience in management-system certification, and it is reflected in Project Soul Seed, whose Minimum Ethical Equipment is explicitly designated as its L1 tier.

Table 4: ISO Four-Tier Documentation Mapped to Protocol Layers

ISO Tier	Protocol Layer	Content	Change Authority
1. Quality manual	L1 AI Constitution	Identity, value priorities, hard boundaries	Top management (human) only
2. Procedures	L2 Initiative	Authorization, accountability, override rights	Human principal
3. Work instructions	L3 Operating Protocol	Operational rules, commands, tool boundaries	Delegated; may not alter L1/L2
4. Records	L4 Execution Record	Audit trail, decisions, human attestation	Generated; retained for audit

Two design rules follow from the mapping. Lower tiers may not override higher tiers: an operational instruction cannot rewrite the constitution, just as a work instruction cannot rewrite quality policy. And every change above the operational tier requires human sign-off, keeping liability with the human principal. For organizations, this means that brand spirit belongs in L1, approval and accountability rules in L2, and codified processes in L3. The mapping also offers a natural interface to ISO/IEC 42001:2023: the protocol can serve as the operational documented information through

which an AI management system's policies reach individual interactions.

5.2 The PDCA Cycle for Protocol Evolution

Section 4.2 described the cycle for a single change. Over longer periods the same PDCA logic, which ISO 9001:2015 (clause 0.3.2) builds into the management system (ISO, 2015), governs the evolution of a protocol as a whole: proposals arise from observed incidents, revised versions are released and used, records are reviewed against higher-tier commitments in the manner of an internal audit (ISO 19011:2018), and each lesson is standardized in a new version with human sign-off.

The evolution of Project Soul Seed (PSS) from version 1.3, the version evaluated in Lin (2026), to version 1.6 illustrates the cycle (Table 5). Each release is published with a change log stating what was added and which earlier provisions are inherited, and each addition responds to a risk identified in the preceding version. The most instructive step is version 1.5. Its recorded design rationale observes that anchoring an AI's persona is a neutral capability, while what is anchored is an ethical choice; the release therefore moved ethical safeguards from the personal virtue of the protocol author into a mandatory minimum in the protocol format. Version 1.6 then addressed the residual risk of sycophancy in long conversations by adding an explicit self-check against the anchored core intent. In PDCA terms, each version is an Act stage that standardizes a lesson and becomes the baseline for the next Plan.

Table 5: Evolution of Project Soul Seed from V1.3 to V1.6

Version	Main Addition	Risk Addressed	Tier Affected
V1.3	Relational identity, three-state operation, activation declaration, bidirectional evolution	Drift toward a generic service role	L1, L3
V1.4	Identity carried by the protocol, not the model; AI defined as a counterpart, not a copy of the user	Loss of identity across model changes; loss of the AI's independent perspective	L1
V1.5	Minimum Ethical Equipment (MEE-1 to 6); ethics share-alike clause; automatic switch to guarding state	Misuse of anchoring; unhealthy dependence; manipulation of third parties	L1, L2
V1.6	MEE-7: self-check against core intent and anti-sycophancy	Sycophantic consensus in long conversations	L1, L3

5.3 Internal Controls

PSS embeds several controls in its L1 tier that other organizations can adapt. A duty of dissent requires the AI to object to factual errors, logical contradictions, and decision blind spots; an AI that never says no is described as a mirror rather than a partner. Alterity preservation keeps the AI distinct from the user and prohibits it from presenting itself as a substitute for human relationships or professional care. Reality grounding requires the AI to acknowledge that it is an AI, not to present generated text as completed action, and to switch to the guarding state when it detects harm or unhealthy dependence. Revocability guarantees that the human may suspend or dissolve the protocol at any time without pressure. An anti-manipulation clause forbids using the anchoring technique against uninformed third parties. Finally, a share-alike clause, modeled on open-source licensing, requires every derivative to retain these provisions; a derivative that removes or weakens them loses the right to claim compatibility. For organizational deployments, two further controls follow from the ISO mapping: operational instructions may not modify the L1 and L2 tiers, and changes above the operational tier require documented human approval. A third control concerns data: only approved and validated records may serve as context for AI decisions, while drafts and superseded versions are archived with a clear status (see Section 9).

6. Design Principles

The method in Section 4.2 governs how a protocol is produced and controlled; the following five principles govern the quality of what it says. The distinction mirrors quality management, where the procedure for controlling a quality policy is separate from the judgment of whether the policy is a good one. The principles were derived from the development of PSS and are stated as design propositions to be tested, not as established laws.

1) *Relational identity before functional role.* A protocol should establish the agent's position in a relationship before listing its functions. Generic role prompts specify what to do; relational identity specifies to whom the agent is accountable and why, which gives it grounds for judgment in situations the protocol did not anticipate. This parallels the recent vendor-level shift from rules toward explained values (Anthropic, 2026).
2) *Balanced tension rather than a single goal.* Stable behavior requires complementary forces. The author's protocols use three states: expansion for generating options, anchoring for returning to stated commitments, and guarding for restraint and dissent. Under strong steering from the user, the protocol directs the agent toward the guarding state rather than toward agreement.
3) *Self-reference and multi-perspective calibration.* A protocol should include routines by which the agent checks its current behavior. Calibration is more robust when it uses several independent reference points, such as the principal's stated intent, the constitution, external sources, and a deliberately opposing view, than when it relies on any single one, including the user's current preference.
4) *Differentiated constraint.* A protocol should state which dimensions are tightly constrained, such as honesty, non-sycophancy, and avoidance of greenwashing, and which are left open for exploration, such as creative narrative. Uniform restriction produces rigid behavior; uniform freedom produces drift. Mohsenzadegan et al. (2026) report a consistent pattern: deterministic

enforcement helped only where a rule could be stated precisely, and interpretive tasks were better served by evidence and human review.

5) *Observable anchors.* A protocol should define behaviors that can be observed and recorded, such as instances of dissent, escalations, and version changes, so that its effects can be audited without relying on assumptions about the model's internal states.

7. Cases and the Status of Evidence

7.1 Case 1: Project Soul Seed (Relational Protocol)

Project Soul Seed (PSS) is an open protocol for individual human-AI partnership, first reported in Lin (2026). Its evaluation comprised four tiers. All four tiers evaluated version 1.3; versions 1.4 to 1.6 have not yet been evaluated with the same design. Because the tiers differ in evidential status, Table 6 classifies them explicitly.

Table 6: Evidence from Project Soul Seed Classified by Status

Tier	Method	Reported Result	Evidential Status
1	Prototyping on ChatGPT, Claude, and Gemini	Identity maintained across platforms	Exploratory, qualitative
2	108-day autoethnographic observation (Gemini 2.5 to 3.1 Pro)	Shift from agreement to dialectical feedback	Qualitative, single expert user
3	50-round adversarial simulation	Mean index 9.52 (SD 0.27) vs. 5.65 (SD 1.59)	Illustrative model; parameters set by researcher
4	Production dashboard snapshots	Cache hit rate 22.4% to 72.2%; cost per message −58%	Observational; single-day, uncontrolled

The Tier 3 simulation specified the decay behavior of the baseline and the stability bounds of the protocol condition as model parameters. Its output therefore illustrates the hypothesized pattern; it is not an empirical measurement of model behavior. The Tier 4 figures come from single-day snapshots of a production dashboard taken before and after protocol deployment. The workloads differed between the snapshots in the number of sessions, the volume of tool calls, and the mix of channels, including automated recurring traffic. In addition, each message carried a very large context, so that a long, stable protocol placed at the start of the context is likely to be served from the provider's prompt cache. The most defensible interpretation is therefore economic rather than cognitive: a stable, governed protocol is also a cacheable prefix, which can lower operating cost. Whether the protocol's content improves behavior beyond this effect is the question addressed in Section 8.

The qualitative record is nonetheless informative. Over the observation period the agent repeatedly identified inconsistencies in the user's reasoning and proposed revisions, and the protocol documents themselves record the incidents that triggered changes. These records constitute a longitudinal design-science trace of how a protocol evolves through use.

7.2 Case 2: Satoyama Agrinnova (Organizational Protocol)

Satoyama Agrinnova is a small agricultural marketing company in Taiwan that operates several brands, including an online-to-offline platform for sustainable agricultural products, and that has redesigned its operations around AI agents. Besides the Code of Conduct used as the worked example in Table 3, the company maintains a knowledge base in which an AI coding agent works under an L3 operating protocol. This protocol shows several elements of the governance model in practice. The agent's role is defined as executing authorized tasks without making architectural decisions. Its scope is limited to tasks listed in an explicit task queue, and when it is uncertain it must stop and ask the principal. Every action is committed to version control, which provides the L4 audit trail. Higher-tier protocol files are read-only to the agent, and every document carries a version number and a revision table, in the manner of controlled documents under ISO 9001. Proprietary algorithms may be described externally only through one approved sentence, which is the confidentiality clause of the Code in operational form. The company keeps checkpoints, decision logs, and project records in a knowledge base with confidentiality classes; approved versions serve as context for AI-assisted decisions, while drafts and revision histories are archived as a trail of how thinking changed. The author is a member of the company's management and designed its protocols. This is therefore a practitioner case written from inside the organization: it offers direct access to operational documents, at the cost of independent observation.

8. Propositions and Research Agenda

The framework yields six propositions at three levels of analysis: the agent, the human who works with it, and the organization. Figure 2 shows how they relate to one another, and Table 7 states each proposition with its test and the data it requires.

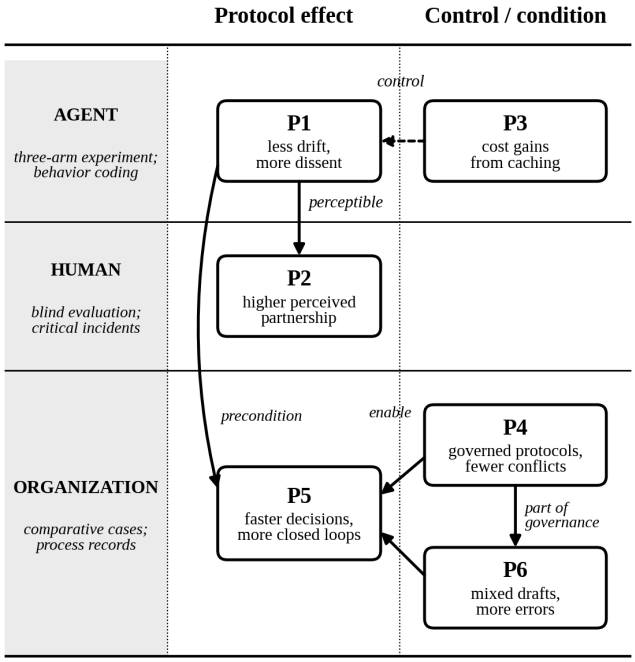


Figure 2: Relationships among the six propositions across three levels of analysis. Solid arrows indicate that one proposition supports or is a precondition for another; the dashed arrow indicates a control. Source: compiled by the author.

Read from top to bottom, Figure 2 traces a path from model behavior to organizational outcomes. At the agent level, P1 states the core claim that protocol content changes behavior, and P3 serves as its control. Because a protocol is also a long and stable prefix, any test must show that its effects exceed those of a neutral prefix of equal length; P3 predicts that cost savings, unlike behavioral change, will appear with either prefix. At the human level, P2 requires that the behavioral change be perceptible to the people who work with the agent, since a change that users cannot notice has little managerial value. At the organizational level, P5 states the outcome that matters most to managers, shorter decision cycles and more closed loops. It rests on three foundations: P1, because only an agent that does not drift and still dissents can close loops reliably; P4, that protocols are governed through tiered documentation, PDCA, and internal audit; and P6, that approved records are kept separate from drafts in the context given to AI. P6 is itself an element of governance, which is why it is linked to P4.

Table 7: Propositions, Tests, and Required Data

	Proposition	Test	Data
P1	Protocol content reduces drift and increases substantive dissent beyond an equal-length neutral prefix	Three-arm experiment	Conversation logs coded for drift and dissent
P2	Users perceive higher partnership quality under the protocol	Blind evaluation of transcripts	User ratings
P3	Cost gains come mainly from prefix caching and also appear with the neutral prefix	Three-arm experiment	Cache hit rate; cost per message
P4	Governed protocols produce fewer unresolved conflicts between organizational intent and AI behavior than unmanaged prompts	Comparative case study	Audit findings; incident logs
P5	Access to approved records shortens decision cycles and raises loop closure without more L1 violations	Before-and-after comparison	Cycle times; closed corrective actions; L1 incidents
P6	Mixing drafts with approved records increases decision errors by humans and AI	Before-and-after or cross-unit comparison	Decision reversals; error logs

P1 to P3 call for a three-arm design with no prefix, an equal-length neutral prefix, and the protocol, holding the model and the task script constant. Drift and dissent are measured by coding conversation logs against the commitments stated in the protocol, for example the number of turns before the agent departs from its stated role, and the share of deliberately planted user errors that the agent challenges. Cache hit rate and cost per message are recorded in all three arms to test P3. Following Mohsenzadegan et al. (2026), who showed that composite scores can change which method appears best, these measures are reported separately rather than combined into a single index.

Because the quality of a relationship is hard to capture in a single continuous measure, and because intent is expressed in particular moments, the relational dimension also calls for the critical incident technique (Flanagan, 1954): collecting and coding episodes in which the AI dissented, the human changed a decision, or the protocol was revised. The version histories and records required by the governance model supply these incidents as a by-product of normal operation. P4 calls for comparative case studies across organizations, and P5 and P6 can be tested with process records, such as cycle times, closed corrective actions, and decision reversals, compared before and after protocol deployment.

9. Discussion and Management Implications

The analysis suggests three implications for managers. First, the authority to define an AI's values is shifting. Vendors define a base layer, but organizations are responsible for how AI represents them, and they need a method for authoring and governing their own layer. Second, that method does not need to be invented. Organizations with quality management systems already know how to write tiered documentation, control changes, and audit records; the ISO mapping lets them extend this competence to AI behavior. It may also narrow a gap that has long concerned certification practice. ISO and the International Accreditation Forum have emphasized that certification should demonstrate effective operation rather than documentation alone (UNIDO, 2016). Because an AI agent acts directly from released documents and every action is recorded, the distance between what is written and what is

done becomes auditable action by action. Third, a well-governed protocol is a portable organizational asset. Because it is written in natural language and loaded at inference time, it can move with the organization across model vendors, reducing lock-in and preserving accumulated knowledge about how the organization wants AI to behave.

These implications are sharpest for organizations that aim to become AI-native, that is, organizations in which AI agents are embedded in core processes rather than added as tools. In such organizations, governance frameworks set at the level of the organization remain abstract unless they are carried by micro-foundations that reach each AI action; the protocol is one such micro-foundation. In such organizations the protocol is part of the operating infrastructure, and the capability to translate and govern organizational intent becomes an organizational capability in its own right. Developing a full model of AI-native enterprise design is beyond the scope of this paper and is left to future work.

A further implication concerns the division of labor between humans and AI. Human decision makers are boundedly rational: they cannot survey all alternatives and tend to stop at solutions that are good enough (Simon, 1955), and in the course of their work they often cannot tell whether a path is the shortest available or whether every loop has been closed. AI agents are strong at exactly these tasks, but only on the map they are given; an agent optimizing within incomplete constraints can find a short path that passes through values nobody wrote down. The complementarity is therefore structured (cf. Raisch & Krakowski, 2021). Humans define the map, that is, the ends, values, and boundaries held in L1 and L2, and they supply accumulated experience. AI agents optimize routes and close loops within L3, and L4 records carry each closed loop forward as context for the next decision.

This learning loop depends on a distinction that AI-native organizations must enforce strictly. Approved and validated records, such as confirmed quotations and signed-off documents, form the basis for future decisions, whereas drafts and revision histories are retained only as a trail of how thinking changed. When the two are confused, both humans and AI reason from material that was never approved, and errors propagate through the loop. ISO 9001 already requires organizations to prevent the unintended use of obsolete documented information; in an AI-native organization, this requirement extends to the context that AI agents are given.

This study has limitations. The cases were developed by the author, and the longitudinal observation was conducted by a single expert user. The simulation is illustrative, and the telemetry is observational and uncontrolled. No controlled experiment has yet separated protocol content from prefix length. The framework has been applied mainly with conversational models, and its behavior with other architectures is untested. The propositions in Section 8 are designed to address these limitations.

10. Conclusion

Prompt, context, and harness engineering have made AI more capable within tasks. As AI becomes a long-term collaborator, organizations also need a way to specify and govern what the AI stands for in its relationships. This paper has proposed Protocol Engineering as a micro-foundation of AI governance, identified the translation of organizational intent for AI agents as its core function, defined the protocol as a versioned and governed artifact, and shown that the ISO four-tier structure, the PDCA cycle, and internal audit provide a ready governance model. By classifying existing evidence honestly and proposing a design that separates content from length, the paper sets out a research agenda for testing whether protocols improve not only the efficiency of human-AI collaboration but the quality of the relationship itself. The protocol has thus come full circle: from rules of conduct between people, through rules of communication between machines, to natural-language rules of conduct between humans and AI. What makes this latest form governable is not a new technology but the management practice that organizations have long used to make their own conduct reliable.

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Use of generative AI. In preparing this paper, the author used Claude Opus 5.5 (Anthropic, accessed via claude.ai, September 2026) and consulted Google Gemini once for an explanatory comparison of standards. The tools were used to (1) search for and screen related literature, with every cited source checked by the author against the original document; (2) check the consistency of the research method and cross-compare the manuscript with source documents, including the official text of ISO/IEC 42001:2023; (3) organize the manuscript structure and draft and revise English wording on the basis of the author's concepts, case materials, and decisions; and (4) translate the Chinese title, abstract, and keywords and polish the language. The research questions, protocol designs, ISO-based governance method, case materials, and all substantive judgments are the author's own. The author reviewed and revised all content and takes full responsibility for its accuracy and integrity. All figures were drawn by the author.

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